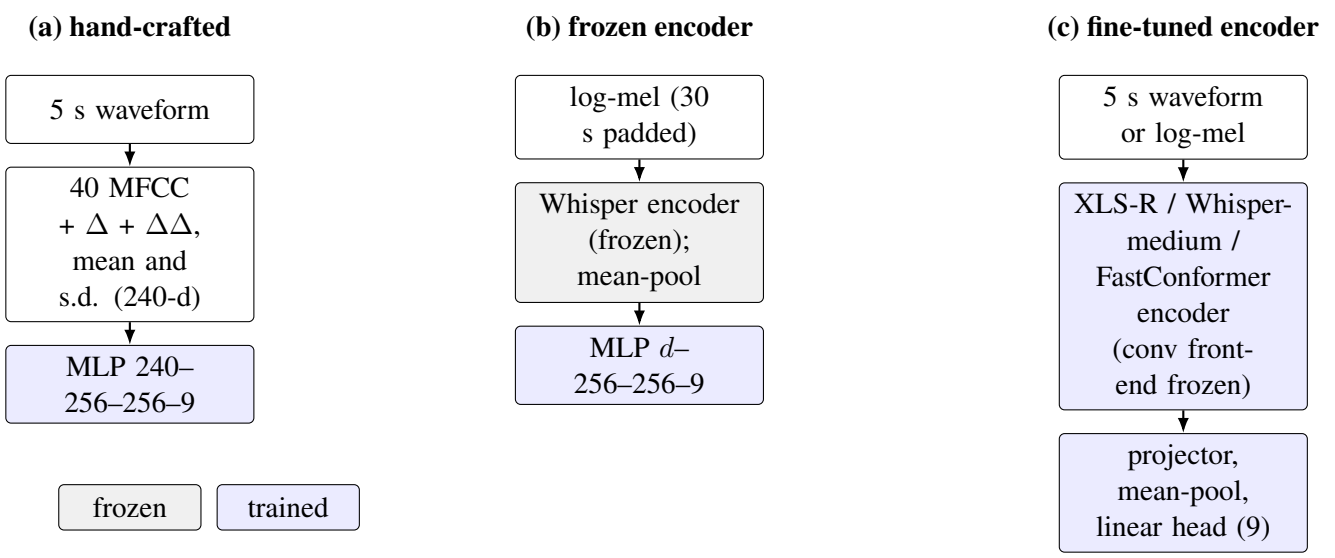
(a) hand-crafted
5 s waveform
40 MFCC + Δ + ΔΔ, mean and s.d. (240-d)
MLP 240–256–256–9
frozen
trained
(b) frozen encoder
log-mel (30 s padded)
Whisper encoder (frozen); mean-pool
MLP d–256–256–9
(c) fine-tuned encoder
5 s waveform or log-mel
XLS-R / Whisper-medium / FastConformer encoder (conv front-end frozen)
projector, mean-pool, linear head (9)